



PREDICTIVE MAINTENANCE FOR TURBOFAN ENGINES USING MACHINE LEARNING

Predicting Remaining Useful Life (RUL) based on NASA C-MAPSS Dataset



THE PROBLEM



- Run-to-Failure:
Too dangerous.
- Scheduled
Maintenance: Too
expensive

THE SOLUTION:

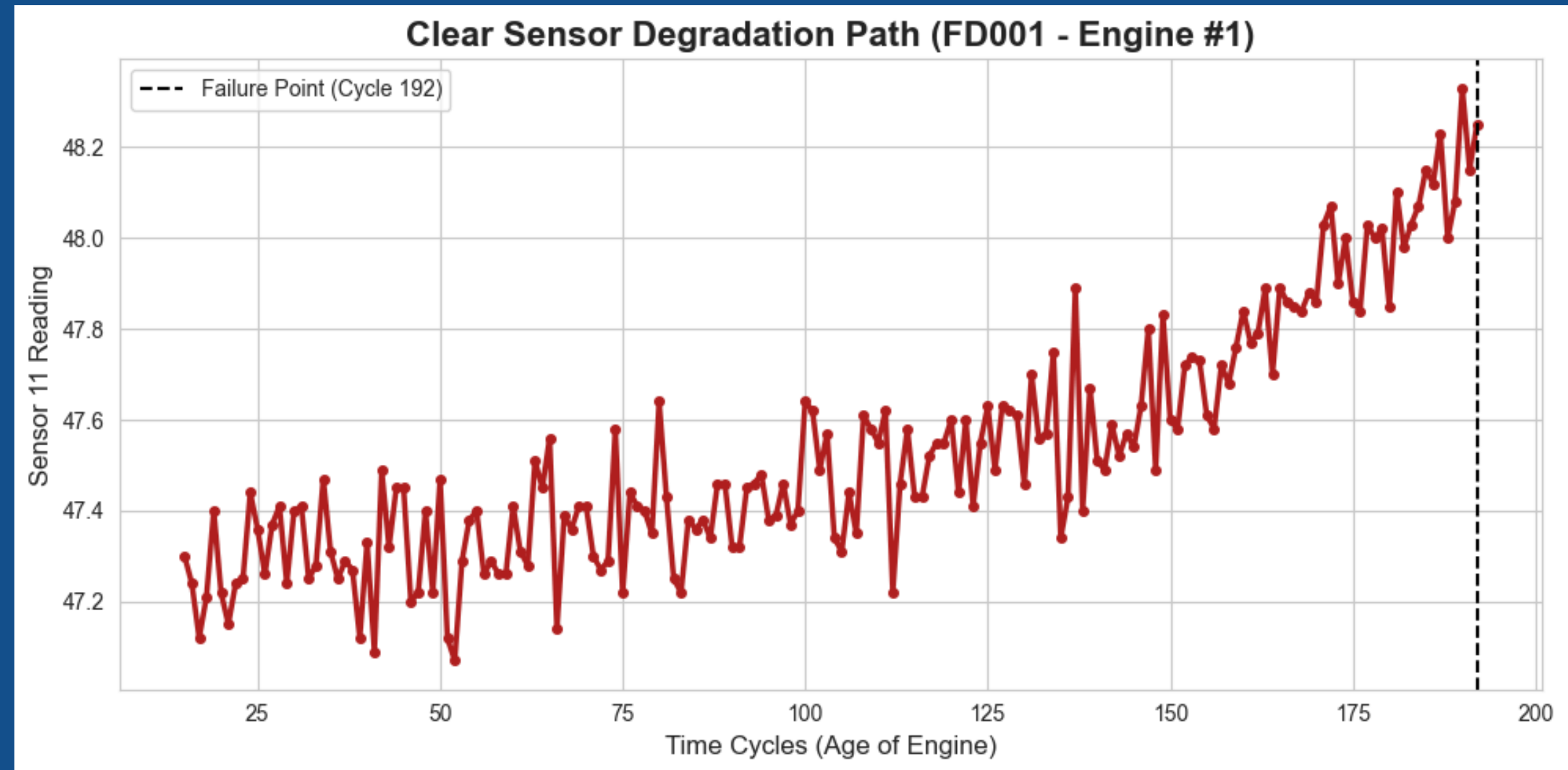
AI-DRIVEN PREDICTIVE MAINTENANCE

- Using Sensor Data to predict the exact time of failure.
- Calculating Remaining Useful Life (RUL).



NASA C-MAPSS DATASET OVERVIEW

- Simulated run-to-failure data for Turbofan engines.
- 21 Sensors recording temperature, pressure, fan speed, etc.



VARYING FAULTS & OPERATING CONDITIONS

FD001:

1 Fault Mode (HPC) + 1
Operating Condition
(Sea Level)

FD004:

2 Fault Modes (HPC &
Fan) + 6 Operating
Conditions.

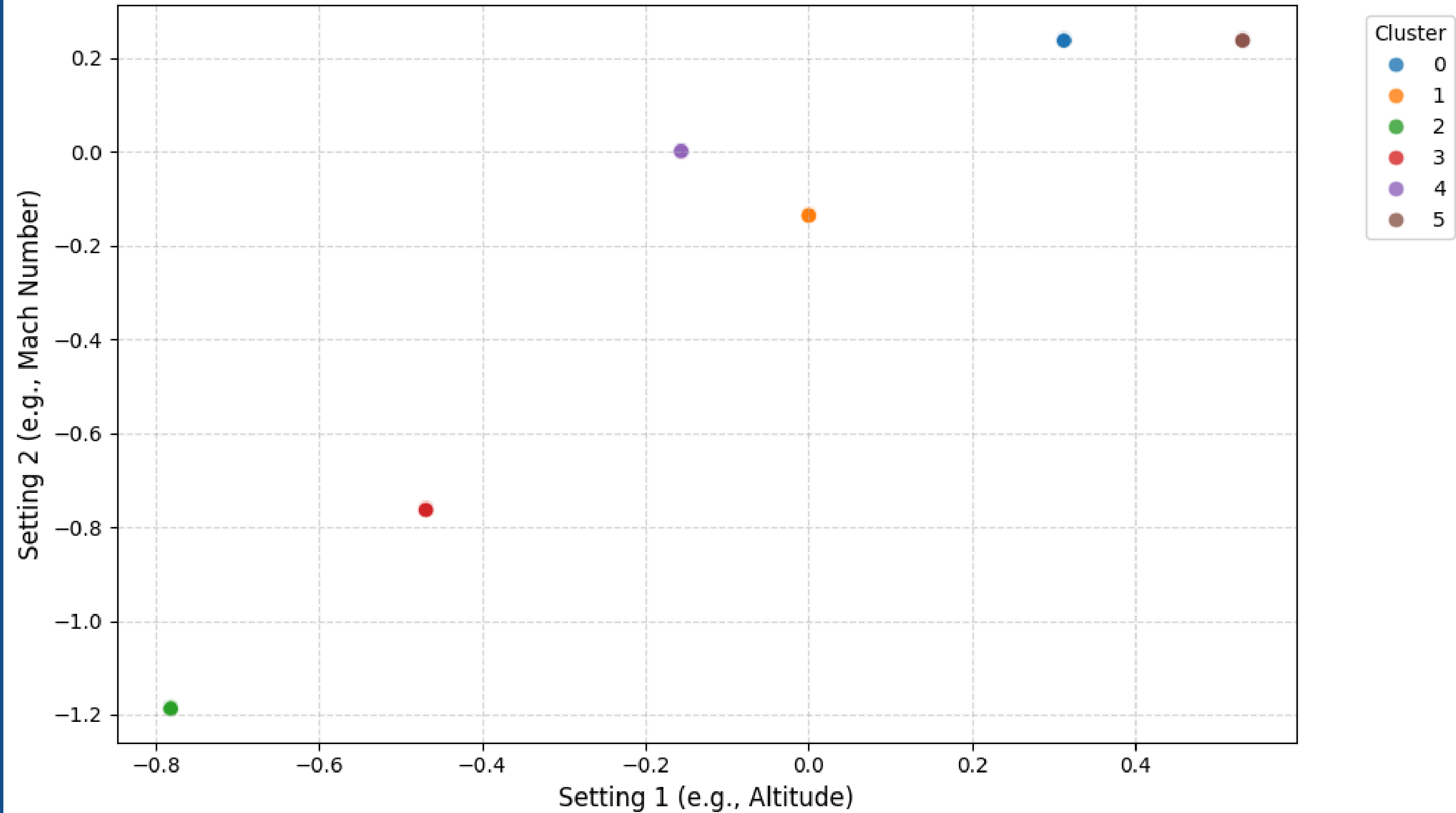
FD002:

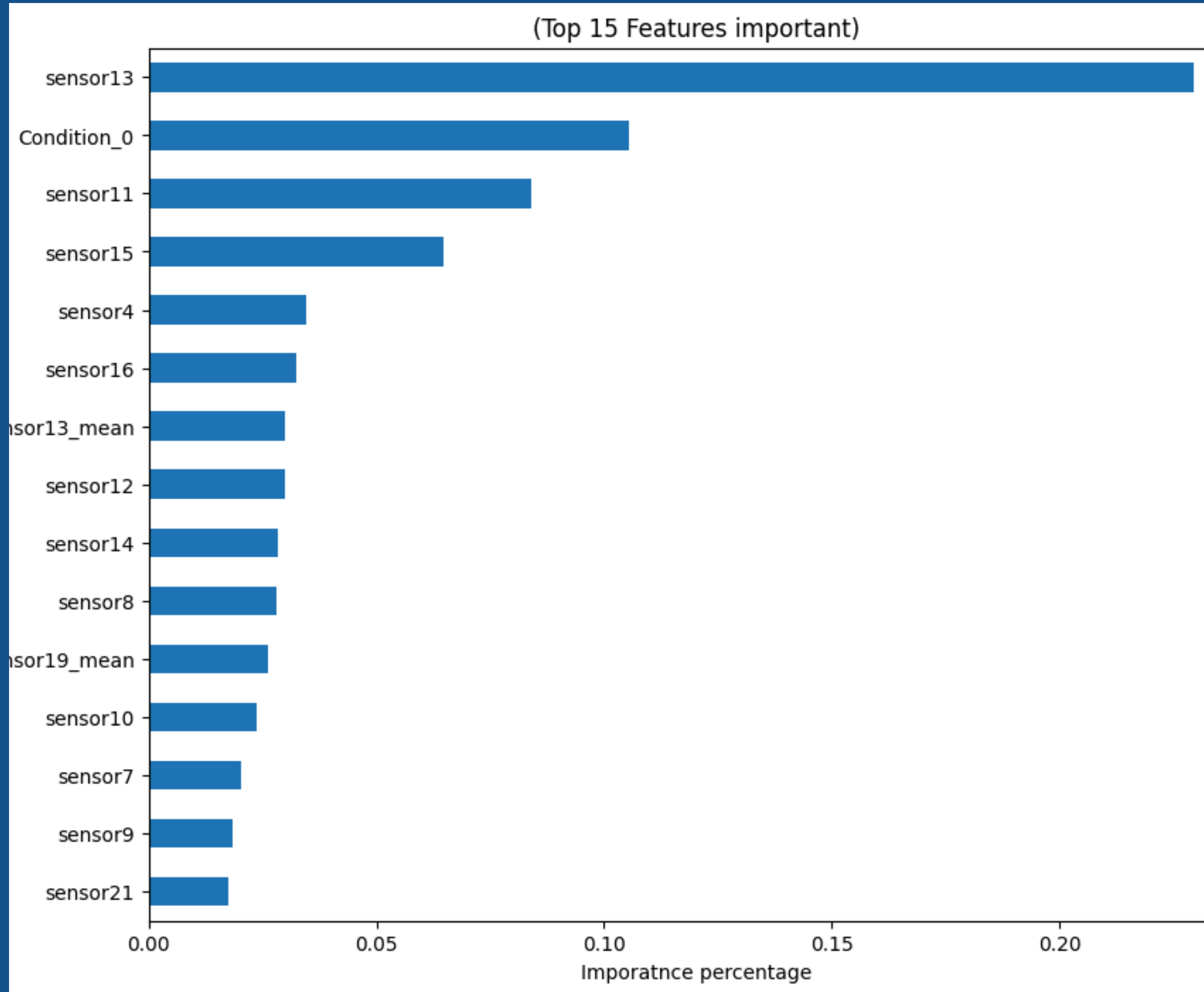
1 Fault Mode (HPC) + 6
Operating Conditions

FD003:

2 Fault Modes (HPC &
Fan) + 1 Operating
Condition (Sea Level)

6 Operating Conditions Detected by K-Means





MODELING STRATEGY

Baseline (FD001):

**Random Forest
Regressor**

Baseline (FD003):

**Random Forest
Regressor**

Baseline (FD002):

**K-Means
Clustering +
XGBoost
Regressor**

Baseline (FD004):

**K-Means
Clustering +
XGBoost
Regressor**

Interactive RUL Predictor Application

Model Settings

Select Dataset (Operating Conditions):

FD002 (1 Fault / 6 Operating ... ▾

Upload Test Data File (.txt)

Drag and drop file here

Limit 200MB per file • TXT

Browse files

test_FD002.txt
5.5MB

Final Result for Engine #1:

Engine Number

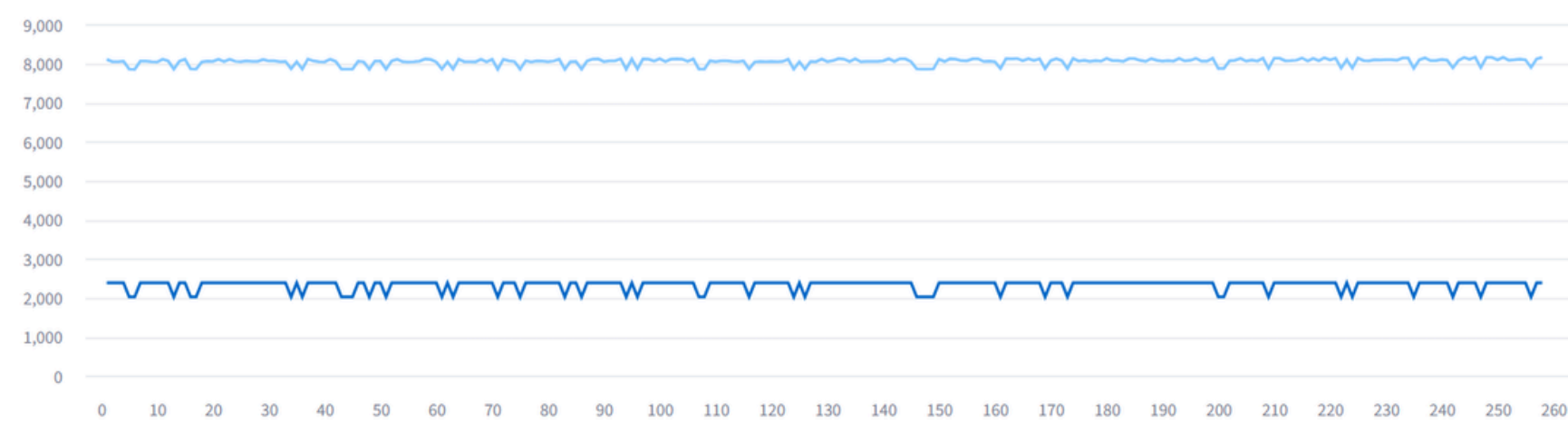
Engine #1

Remaining Useful Life (RUL)

8 Cycles

↓ - CRITICAL! Maintenance Required

Sensor Degradation History for Engine #1



Deploy ⋮

Model Settings

Select Dataset (Operating Conditions):

FD004 (2 Faults / 6 Operatin... ▾

Upload Test Data File (.txt)

Drag and drop file here

Limit 200MB per file • TXT

Browse files

test_FD004.txt
6.6MB

Final Result for Engine #4:

Engine Number

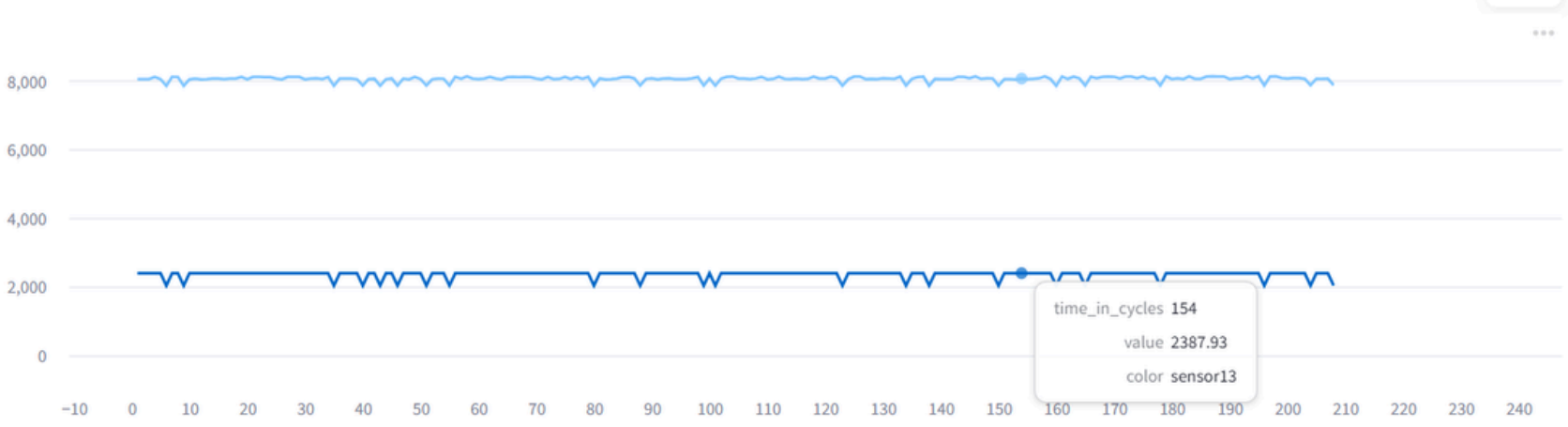
Engine #4

Remaining Useful Life (RUL)

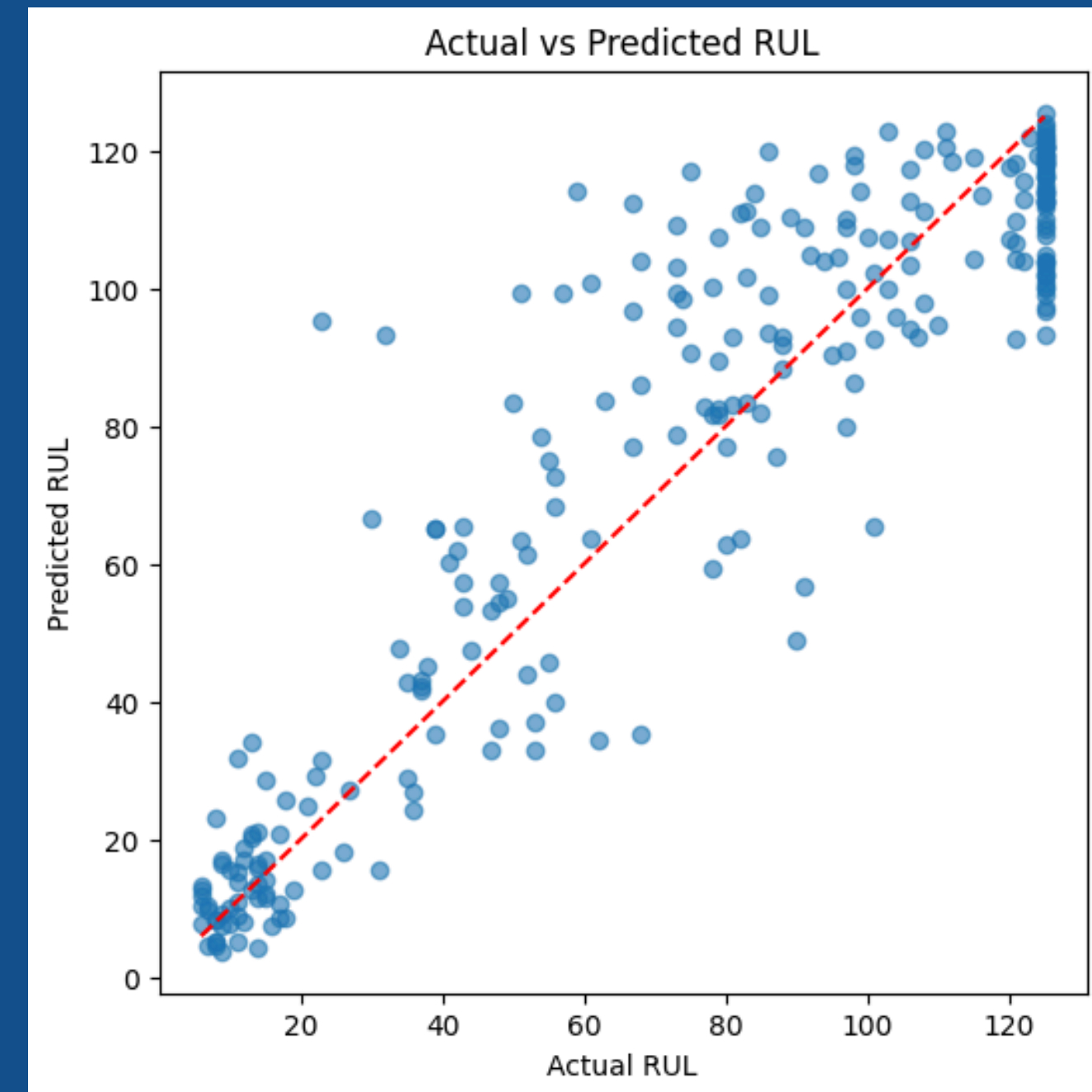
113 Cycles

↑ + Healthy / Safe Condition

Sensor Degradation History for Engine #4



CONCLUSION & IMPACT



- Successfully predicted engine failure.
- Transitioned from theoretical models to a usable application



THANK YOU

Sara Abdel Hakeem
Rowaida Abdallah